Lecture 12: Mesh Current Analysis

EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current, mesh current 1 voltage supermesh

Review: node voltage analysis

The procedure

- Ground; node voltages $\mathbf{e}$
- KVL automatic: $\mathbf{v} = A^T \mathbf{e}$
- KCL at non-ground nodes
- $Y_n \mathbf{e} = \mathbf{i}_n$

Review: node voltage analysis

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Voltage sources

- To the ground: e_k known
- Between two nodes: supernode
- Dependent: one more constraint

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The opposite?

Motivation

Node voltages

- Unknowns: $n - 1$ node voltages
- KVL automatic
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- Unknowns: $\ell = b - n + 1$ mesh currents

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Mesh currents

- Unknowns: $\ell = b - n + 1$ mesh currents
- KCL automatic: $\mathbf{i} = M^T \mathbf{i}_M$
- Write KVL
- Needs a planar circuit

Roadmap









Voltage sources | **Current sources** | **Dependent sources** | **Node or mesh?**

1 — 2 — 3 — 4

the procedure, on an example

matrix form

on one mesh only

between two meshes

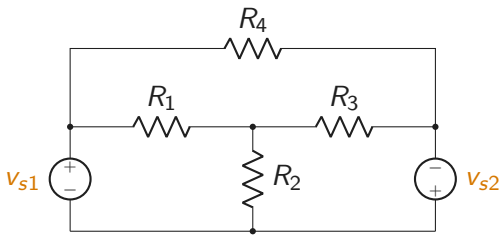
a dependent current source

a dependent voltage source

count the equations

Example 12.1: mesh current analysis step by step

For now: only resistors and independent voltage sources

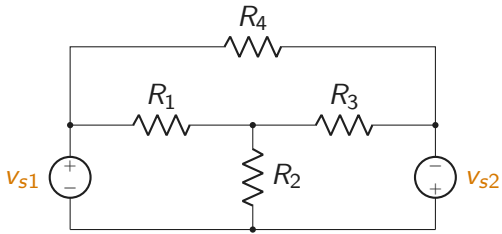


Goal: every branch current and voltage

Four steps:

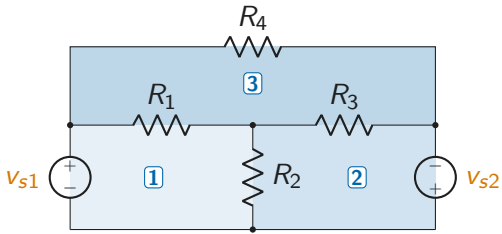
1. Find the inner meshes
2. Assign the mesh currents
3. Resistor voltages with mesh currents
4. KVL around every inner mesh

Step 1: find the inner meshes



Planar: no crossing branches

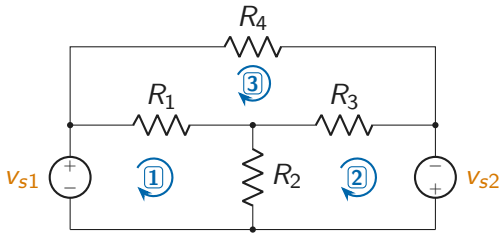
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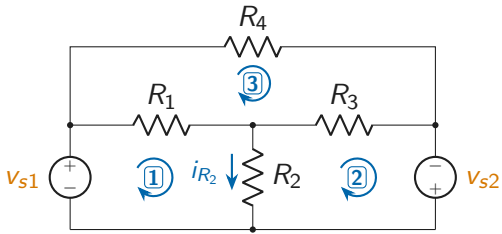
Inner meshes: the windows

Step 2: assign the mesh currents



Unknowns, clockwise: i_{M1}, i_{M2}, i_{M3}

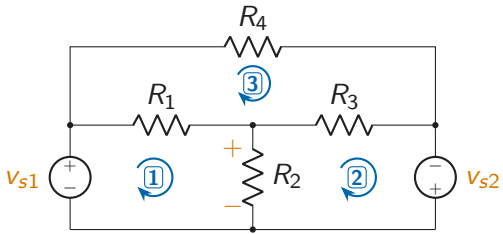
Step 2: assign the mesh currents



Unknowns, clockwise: i_{M1}, i_{M2}, i_{M3}

$i_{R_2} = i_{M1} - i_{M2}$

Step 3: write the resistor voltages with mesh currents



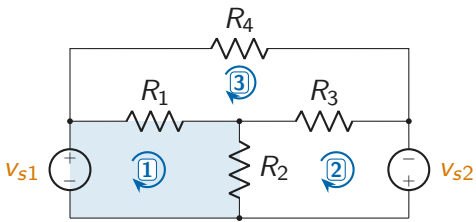
R_2, along mesh 1:

$v = R_2(i_{M1} - i_{M2})$

R_4: $v = R_4\, i_{M3}$

R (own − other)

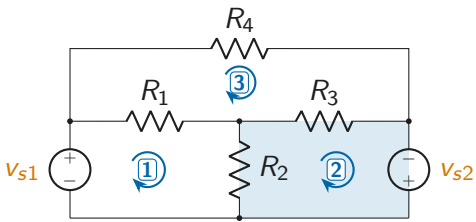
Step 4: KVL around every inner mesh



Mesh 1:

$$-v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0$$

Step 4: KVL around every inner mesh



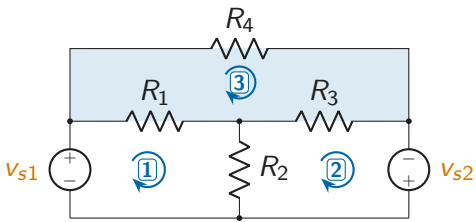
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Mesh 2:

$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

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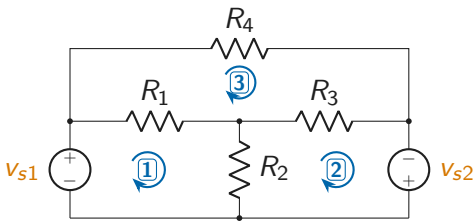
Mesh 2:

$$R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0$$

Mesh 3:

$$R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0$$

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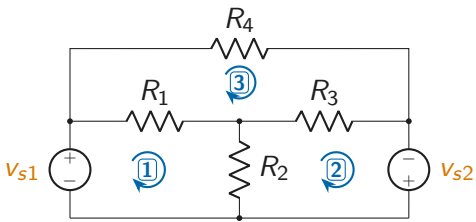
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Source met $-$ to $+$: minus

Step 4: KVL around every inner mesh



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Source met $-$ to $+$: minus

3 equations, 3 unknowns

Roadmap









Voltage sources | **Current sources** | **Dependent sources** | **Node or mesh?**

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the procedure, on an example ✓

matrix form

on one mesh only

between two meshes

a dependent current source

a dependent voltage source

count the equations

Example 12.1 in matrix form

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\begin{aligned}
(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\
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Z_M: mesh impedance matrix

Example 12.1 in matrix form

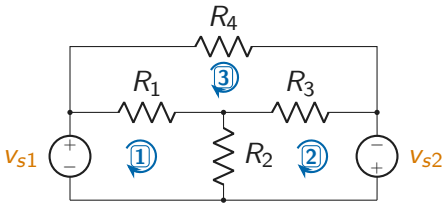
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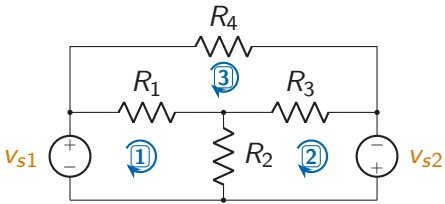
$\mathbf{v}_M$: source voltages rising along each mesh

Writing Z_M directly from the circuit



$$\underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{Z_M} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\mathbf{v}_M}$$

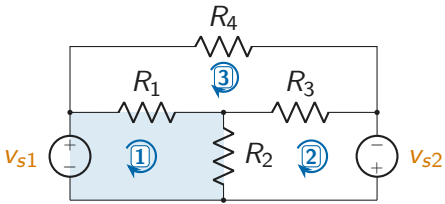
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Diagonal (k, k): $+\sum R$ around mesh k

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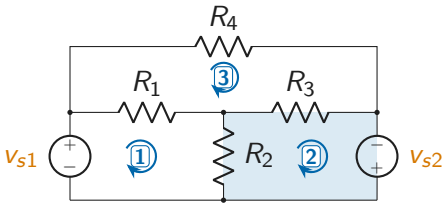
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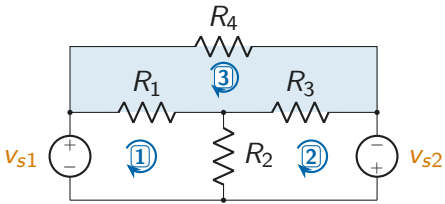
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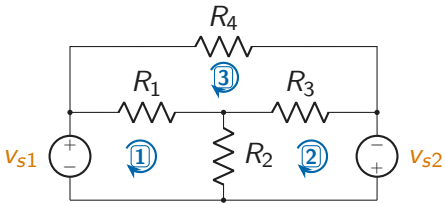
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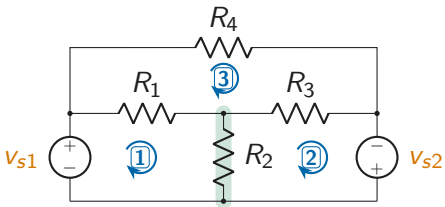


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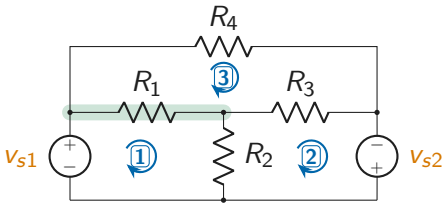


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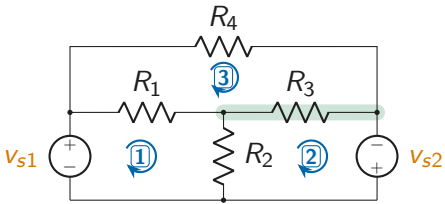


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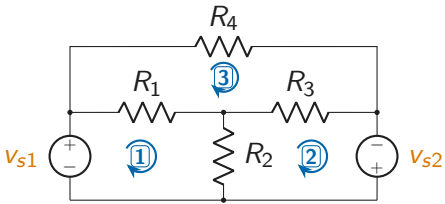


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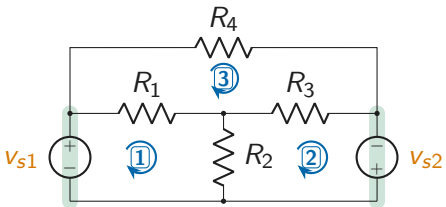
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Right side k: $\sum v_s$ rising along mesh k

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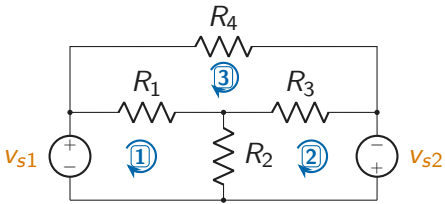
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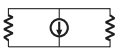
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Same as KVL ✓ Z_M is symmetric

Roadmap









Voltage sources | **Current sources** | **Dependent sources** | **Node or mesh?**

✓ | 2 | 3 | 4

the procedure, on an example ✓

matrix form ✓

on one mesh only

between two meshes

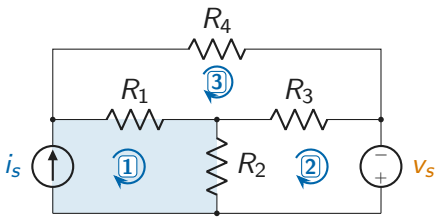
a dependent current source

a dependent voltage source

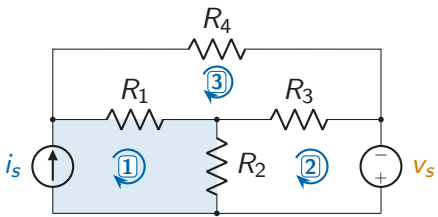
count the equations

Example 12.2: a current source on one mesh only

$i_{M1} = i_s$: known



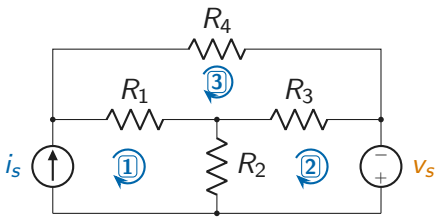
Example 12.2: a current source on one mesh only



$i_{M1} = i_s$: known

No KVL around mesh 1. Unknowns: i_{M2}, i_{M3}

Example 12.2: a current source on one mesh only

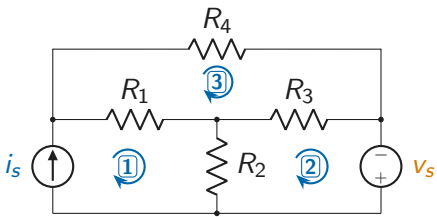


$i_{M1} = i_s$: known

No KVL around mesh 1. Unknowns: i_{M2}, i_{M3}

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Example 12.2: a current source on one mesh only



$i_{M1} = i_s$: known

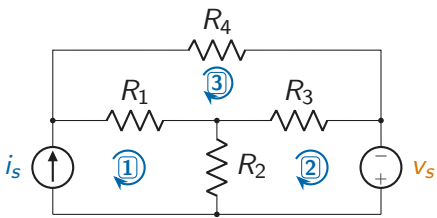
No KVL around mesh 1. Unknowns: i_{M2}, i_{M3}

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

Example 12.2: a current source on one mesh only



$i_{M1} = i_s$: known

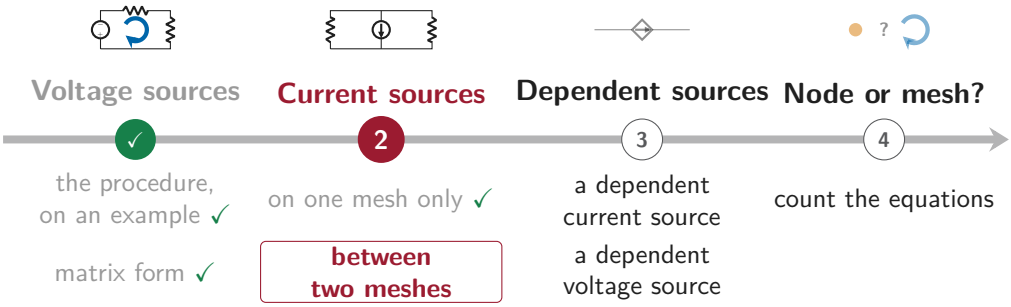
No KVL around mesh 1. Unknowns: i_{M2}, i_{M3}

Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

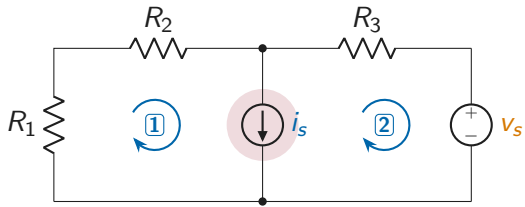
Mesh 3:
$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

Roadmap

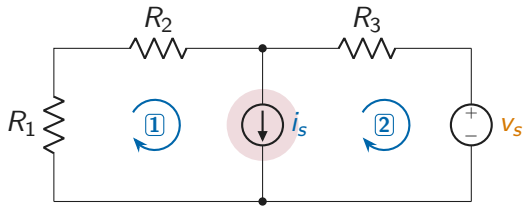

Voltage sources
Current sources
Dependent sources
Node or mesh?
✓
2
3
4
the procedure, on an example ✓
matrix form ✓
on one mesh only ✓
between two meshes
a dependent current source
a dependent voltage source
count the equations

Example 12.3: a current source between two meshes



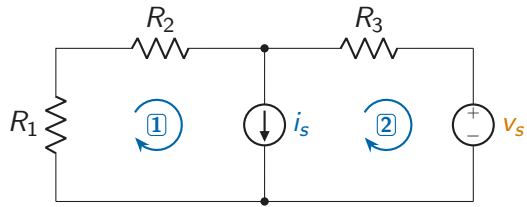
- Voltage across i_s: ?

Example 12.3: a current source between two meshes



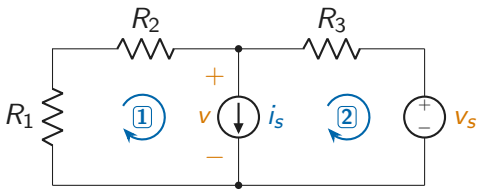
- Voltage across i_s: ?
- Not given by i_{M1}, i_{M2}: a current source fixes i, not v

Example 12.3: a current source between two meshes



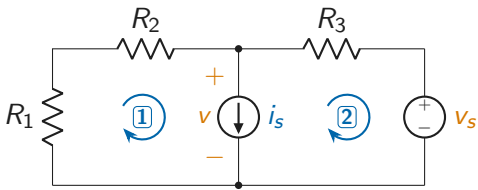
- Voltage across i_s: ?
- Not given by i_{M1}, i_{M2}: a current source fixes i, not v
- Method I: its voltage as an unknown Method II: supermesh

Method I: the voltage as an unknown



New unknown: v

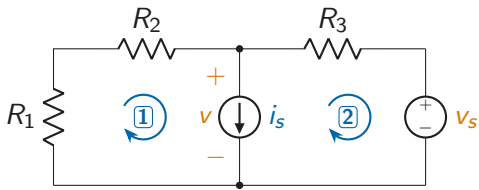
Method I: the voltage as an unknown



Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

New unknown: v

Method I: the voltage as an unknown

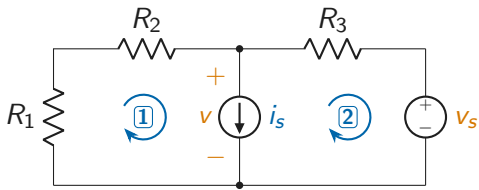


Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

New unknown: v

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Method I: the voltage as an unknown



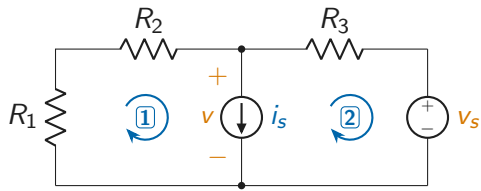
Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

New unknown: v

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Method I: the voltage as an unknown

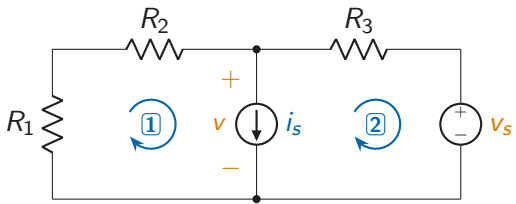


$$\begin{bmatrix} R_1+R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

New unknown: v

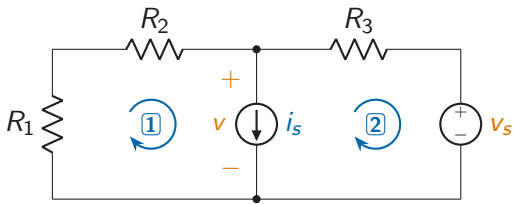
3 unknowns, 3 equations

Method II: the supermesh



Can we get rid of v?

Method II: the supermesh

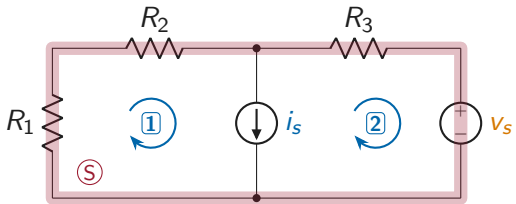


Can we get rid of v?

Add KVL of meshes 1 and 2:

v cancels

Method II: the supermesh

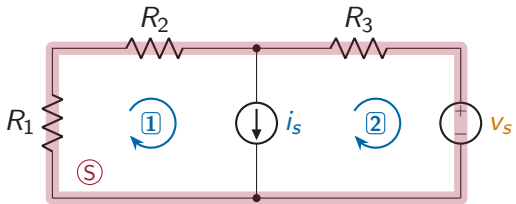


Can we get rid of v? = KVL around the outer loop Ⓢ

Add KVL of meshes 1 and 2:

v cancels

Method II: the supermesh



Can we get rid of v?

Add KVL of meshes 1 and 2:

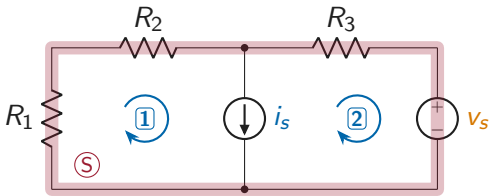
v cancels

$=$ KVL around the outer loop (S)

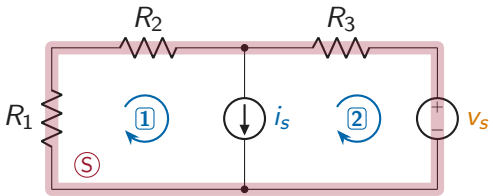
Supermesh

Two meshes as one, skipping the current source

Method II: the equations

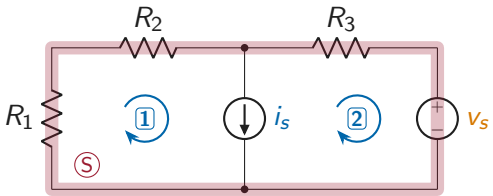


Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

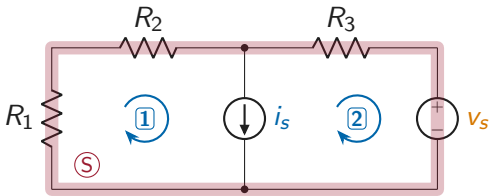
Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Method II: the equations

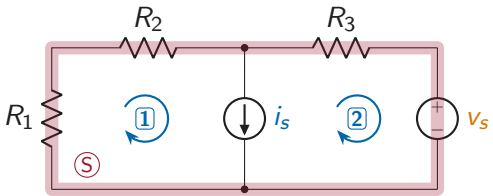


Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

Method II: the equations



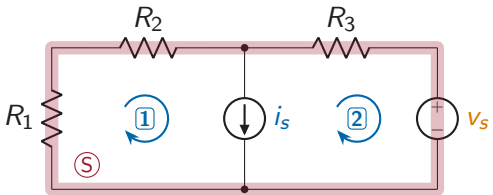
Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

2 unknowns (one fewer)

Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

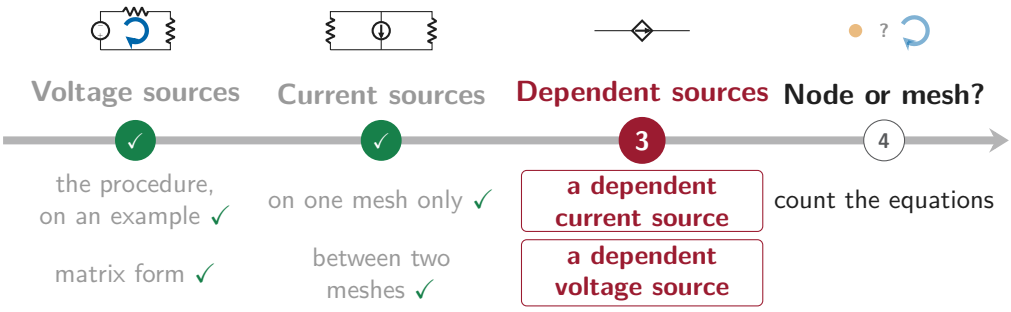
Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

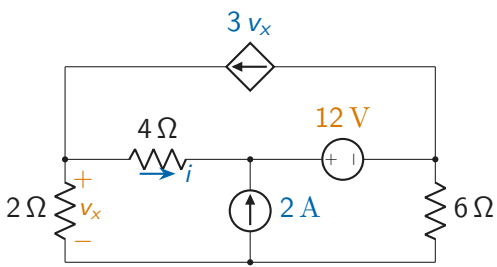
2 unknowns (one fewer)

Not symmetric

Roadmap

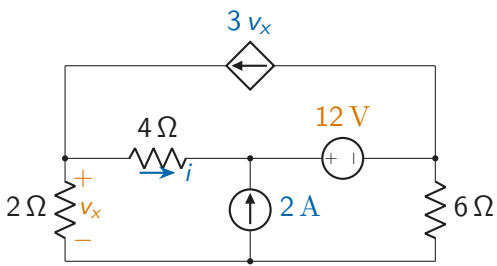

Voltage sources
Current sources
Dependent sources
Node or mesh?
?
✓
✓
3
4
the procedure, on an example ✓
matrix form ✓
on one mesh only ✓
between two meshes ✓
a dependent current source
a dependent voltage source
count the equations

Example 12.4: a dependent current source



Find i of the 4 Ω.

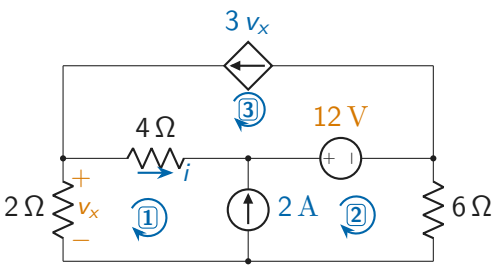
Example 12.4: a dependent current source



Find i of the 4 Ω.

Treat $3\,v_x$ as a current source

Example 12.4: a dependent current source

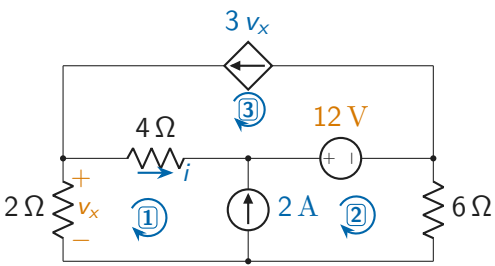


Find i of the 4 Ω.

Treat $3\,v_x$ as a current source

Unknowns: i_{M1}, i_{M2}, i_{M3}

Example 12.4: a dependent current source



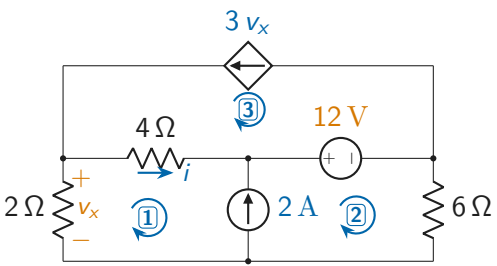
Find i of the 4 Ω.

Treat $3\,v_x$ as a current source

Unknowns: i_{M1}, i_{M2}, i_{M3}

2 A between meshes 1, 2: supermesh

Example 12.4: a dependent current source



Find i of the 4 Ω.

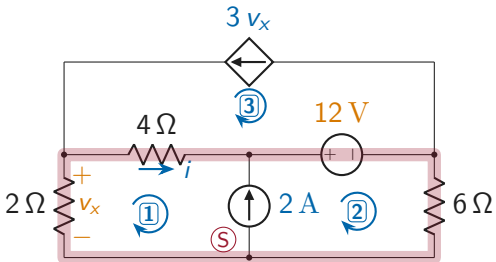
Treat 3 v_x as a current source

Unknowns: i_{M1}, i_{M2}, i_{M3}

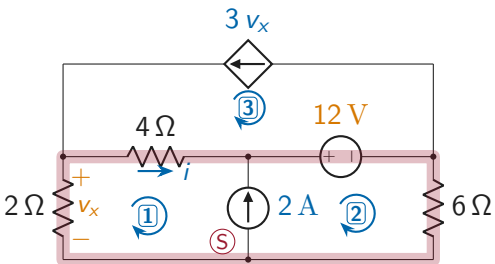
2 A between meshes 1, 2: supermesh

3 v_x on mesh 3 only: fixes i_{M3}

Example 12.4: the equations



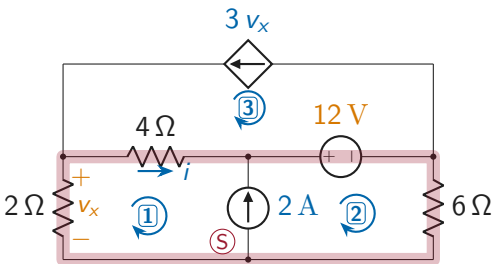
Example 12.4: the equations



Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

Example 12.4: the equations

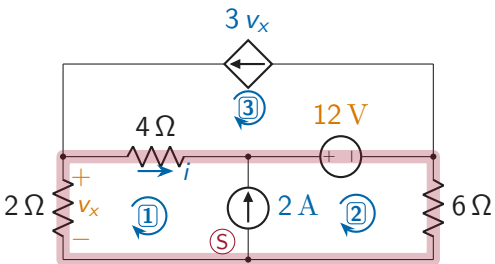


Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

2 A **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Example 12.4: the equations



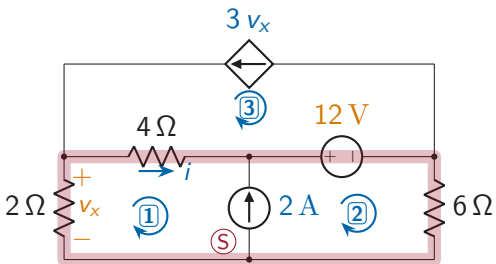
Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

2 A **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Dependent source: $i_{M3} = -3\,v_x$

Example 12.4: the equations



Supermesh:

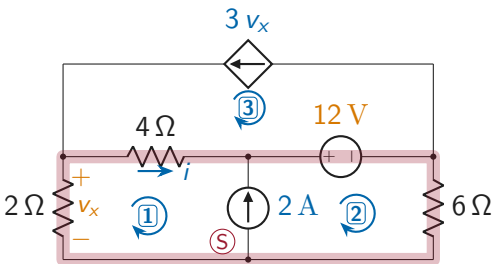
$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$

$2\,\mathrm{A}$ **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Dependent source: $i_{M3} = -3\,v_x$

$v_x = ?$

Example 12.4: the equations



Supermesh:

$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$

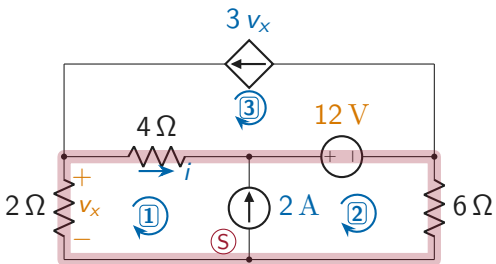
$2\,\mathrm{A}$ **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Dependent source: $i_{M3} = -3\,v_x$

$v_x = ?$

$v_x = -2\,i_{M1}$

Example 12.4: the equations



Supermesh:

$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$

$2\,\mathrm{A}$ **source:** $i_{M2} - i_{M1} = 2 \quad (2)$

Dependent source: $i_{M3} = -3\,v_x$

$v_x = ?$

$v_x = -2\,i_{M1}$

Constraint: $i_{M3} = 6\,i_{M1} \quad (3)$

Example 12.4: matrix form

(1), (2), (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

Example 12.4: matrix form

(1), (2), (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

$$i_{M1} = 2\,\mathrm{A}, \quad i_{M2} = 4\,\mathrm{A}, \quad i_{M3} = 12\,\mathrm{A}$$

Example 12.4: matrix form

(1), (2), (3):

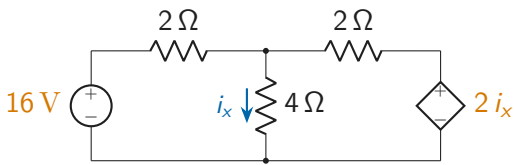
$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

$$i_{M1} = 2\,\mathrm{A}, \quad i_{M2} = 4\,\mathrm{A}, \quad i_{M3} = 12\,\mathrm{A}$$

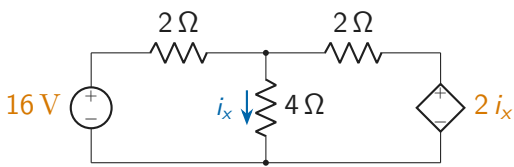
$$i = i_{M1} - i_{M3} = 2 - 12 = -10\,\mathrm{A}$$

Example 12.5: a dependent voltage source

Find i_x.



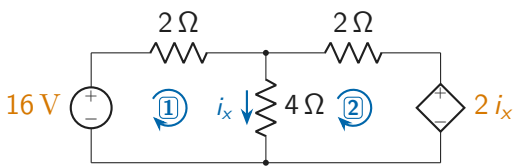
Example 12.5: a dependent voltage source



Find i_x.

Treat $2\,i_x$ as a voltage source

Example 12.5: a dependent voltage source

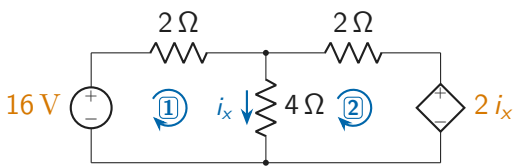


Find i_x.

Treat $2\,i_x$ as a voltage source

Mesh 1: $-16 + 2\,i_{M1} + 4(i_{M1} - i_{M2}) = 0$

Example 12.5: a dependent voltage source



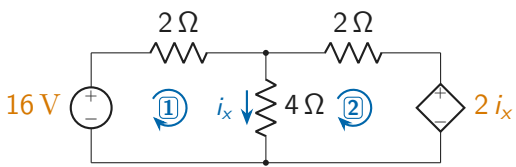
Find i_x.

Treat $2\,i_x$ as a voltage source

Mesh 1: $-16 + 2\,i_{M1} + 4(i_{M1} - i_{M2}) = 0$

Mesh 2: $4(i_{M2} - i_{M1}) + 2\,i_{M2} + 2\,i_x = 0$

Example 12.5: a dependent voltage source



Find i_x.

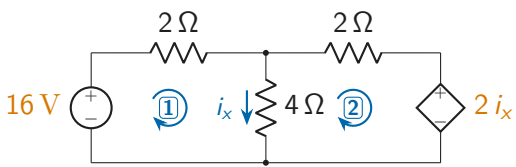
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$i_x = ?$

Example 12.5: a dependent voltage source



Find i_x.

Treat $2\,i_x$ as a voltage source

Mesh 1: $-16 + 2\,i_{M1} + 4(i_{M1} - i_{M2}) = 0$

Mesh 2: $4(i_{M2} - i_{M1}) + 2\,i_{M2} + 2\,i_x = 0$

$i_x = ?$

$i_x = i_{M1} - i_{M2}$

Example 12.5: matrix form

First plug $i_x = i_{M1} - i_{M2}$ into mesh 2:

$$4(i_{M2} - i_{M1}) + 2\,i_{M2} + 2(i_{M1} - i_{M2}) = 0$$

Example 12.5: matrix form

First plug $i_x = i_{M1} - i_{M2}$ into mesh 2:

$$4(i_{M2} - i_{M1}) + 2\,i_{M2} + 2(i_{M1} - i_{M2}) = 0 \qquad \Rightarrow \qquad -2\,i_{M1} + 4\,i_{M2} = 0$$

Example 12.5: matrix form

First plug $i_x = i_{M1} - i_{M2}$ into mesh 2:

$$4(i_{M2} - i_{M1}) + 2\,i_{M2} + 2(i_{M1} - i_{M2}) = 0 \qquad \Rightarrow \qquad -2\,i_{M1} + 4\,i_{M2} = 0$$

$$\begin{bmatrix} 6 & -4 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} 16 \\ 0 \end{bmatrix}$$

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Example 12.5: matrix form

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$$4(i_{M2} - i_{M1}) + 2\,i_{M2} + 2(i_{M1} - i_{M2}) = 0 \qquad \Rightarrow \qquad -2\,i_{M1} + 4\,i_{M2} = 0$$

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$$\boxed{i_x = i_{M1} - i_{M2} = 4 - 2 = 2\,\mathrm{A}}$$

Example 12.5: matrix form

First plug $i_x = i_{M1} - i_{M2}$ into mesh 2:

$$4(i_{M2} - i_{M1}) + 2\,i_{M2} + 2(i_{M1} - i_{M2}) = 0 \qquad \Rightarrow \qquad -2\,i_{M1} + 4\,i_{M2} = 0$$

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$$i_x = i_{M1} - i_{M2} = 4 - 2 = 2\,\mathrm{A}$$

$-4 \neq -2$: not symmetric

Dependent sources

- Like an independent source of the same kind

Dependent sources

- Like an independent source of the same kind
- Control variable in mesh currents

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- Like an independent source of the same kind
- Control variable in mesh currents
- First plug it in, then form the matrix

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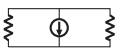
Remark

Dependent current source on mesh k only:

no KVL around mesh k; write the constraint instead

Roadmap









Voltage sources ✓

the procedure, on an example ✓

matrix form ✓

Current sources ✓

on one mesh only ✓

between two meshes ✓

Dependent sources ✓

a dependent current source ✓

a dependent voltage source ✓

Node or mesh? 4

count the equations

Node or mesh?

Example 12.4: which needs fewer equations?

Node or mesh?

Example 12.4: which needs fewer equations?

Mesh currents

$\ell = 3$, 2 current sources

$3 - 2 = 1$ equation

Node or mesh?

Example 12.4: which needs fewer equations?

Mesh currents

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$3 - 2 = 1$ equation

Node voltages

$n - 1 = 3$, 1 voltage source

$3 - 1 = 2$ equations

Node or mesh?

Example 12.4: which needs fewer equations?

Mesh currents

$\ell = 3$, 2 current sources

$3 - 2 = 1$ equation

Node voltages

$n - 1 = 3$, 1 voltage source

$3 - 1 = 2$ equations

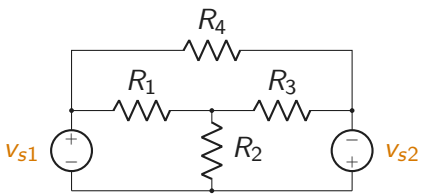
Tip. Count first. Mesh needs a planar circuit.

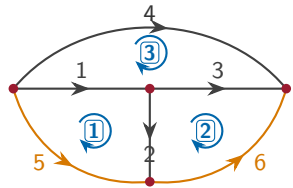
Summary

In the circuit	What to do
resistors, voltage sources	$Z_M \mathbf{i}_M = \mathbf{v}_M$, by the rules; symmetric
current source on one mesh	i_{Mk} known; no KVL around mesh k
current source between meshes	Method I or supermesh, plus constraint
dependent source	control in mesh currents: constraint
node or mesh?	count; mesh needs planar

Then $\mathbf{i} = M^T \mathbf{i}_M$ and the element laws

Z_M from the graph: the mesh matrix

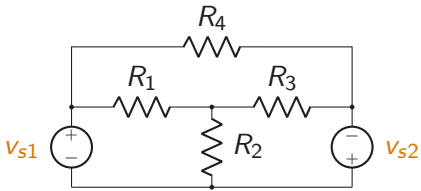


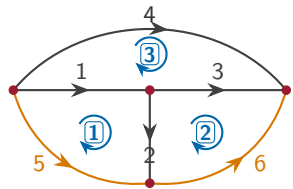


1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} (+ to −)

$M = ?$

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} ($+$ to $-$)

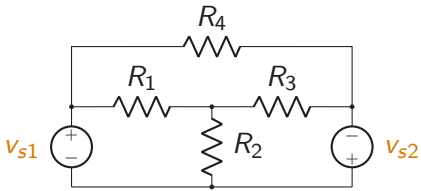
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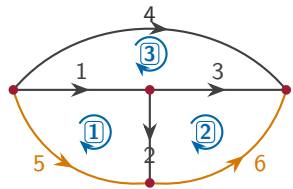
One row per inner mesh

$+1$ along, -1 against

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} ($+$ to $-$)

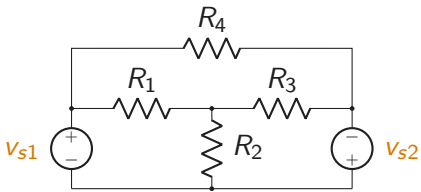
$M = ?$

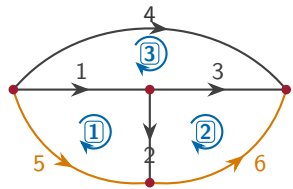
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$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} (+ to −)

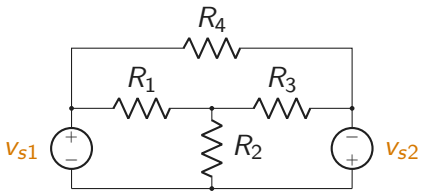
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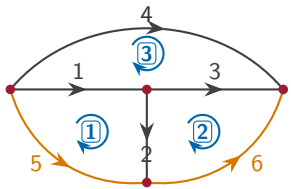
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$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2} (+ to −)

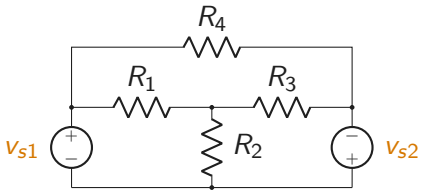
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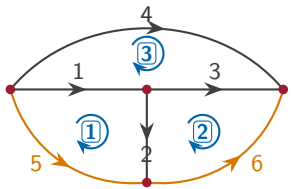
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Z_M from the graph: the mesh matrix





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One row per inner mesh

$+1$ along, -1 against

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M_R: resistors M_S: sources

Z_M from the graph: the general result

Resistors first, then voltage sources: $M = [\, M_R \quad M_S \,]$

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$$M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

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$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

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Resistors first, then voltage sources: $M = [\, M_R \;\; M_S \,]$

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$$\boxed{Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S}$$

$Z_M^T = Z_M$: always symmetric

Z_M from the graph: Example 12.1

From M: M_R = columns 1–4, M_S = columns 5, 6

$$M_R\,R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1

From M: M_R = columns 1–4, M_S = columns 5, 6

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1

From M: M_R = columns 1–4, M_S = columns 5, 6

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Z_M from the graph: Example 12.1

From M: M_R = columns 1–4, M_S = columns 5, 6

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

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$$-M_S \mathbf{v}_S = -\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}$$

Z_M from the graph: Example 12.1

From M: M_R = columns 1–4, M_S = columns 5, 6

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Z_M from the graph: Example 12.1

From M: M_R = columns 1–4, M_S = columns 5, 6

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix} = Z_M \quad \checkmark$$

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The rules = this product